Replication of *The Efficacy of Seven Gratitude Interventions for Promoting Subjective Well-Being* by Dang et al. (2025, *Journal of Positive Psychology*)

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Introduction

[No abstract is needed.] Each replication project will have a straightforward, no frills report of the **study and results.** These reports will be publicly available as supplementary material for the aggregate report(s) of the project as a whole. Also, to maximize project integrity, the intro and methods will be written and critiqued in advance of data collection. **Introductions can be just 1-2 paragraphs clarifying the main idea of the original study, the target finding for replication, and any other essential information.** You will likely contact the authors of the original study, so this should be professional and forward facing. It will NOT have a literature review – that is in the original publication. You can write both the introduction and the methods in past tense.

Experimental Paradigm

The paper by Dang et al. (2026) sought to test 7 gratitude interventions that are used in positive psychology to improve subjective well-being. These interventions included intrapersonal, interpersonal, and gratitude-to-God interventions. A key strength of this experiment is that most studies in the past used different measures of subjective well-being and only compared a few of the interventions against each other at one time which made it hard to make comparisons across interventions. By including a comprehensive battery of interventions and standardized outcome measures of state gratitude, state positive/negative affect, and life satisfaction, this experiment allows for a direct comparison between gratitude interventions and can inform recommendations for everyday people to incorporate into their lives.

Relevance to Research Interests. In my personal research, I am interested in testing low-cost, accessible, and portable single-session interventions that can improve subjective well-being as well as uncover the mechanisms which underlie these changes to subjective well-being, particularly in student populations. Students represent a unique population with higher rates of depression and anxiety as well as potentially limited financial resources to seek help. My research seeks to thus find preventative practices that can be used to buffer against the stressors of big life changes, coursework, and future career uncertainty. This experiment by Dang et al. (2025) has found that privately writing a gratitude letter that does not get shared with anyone has a sizeable impact on subjective well-being across a variety of outcomes including increasing gratitude and positive affect, as well as reducing negative affect. Interventions such as these are precisely the kinds of interventions I hope to design, expand upon, and test due to its brevity and financial accessibility.

Description of Stimuli and Procedures/Challenges. The original Dang et al. (2025) experiment tested 7 different interventions; however, I am most interested in the effect that the gratitude letter (vs. writing an events list active control) has on state positive affect. There was a larger effect of the gratitude intervention when compared to only measuring state positive affect in the control (Hedges' $g = .61$); however, as Haoyu suggested, having an active control that would be able to control for the effects that writing may have on positive affect would

strengthen our ability to conclude that it was the gratitude aspect of writing that letter than lead to increases in happiness. This effect (gratitude letter vs. writing a list of events) was smaller but still quite large (Hedges' $g = .41$).

In the interest of replication, I will use the same instructions that were provided in the original study (Cunha et al., 2019; Dang et al., 2025; Emmons & McCullough, 2003; Walsh et al., 2022). Participants will be randomly assigned to two conditions and see the following directions:

- **Control Group: Events List (Cunha et al. 2019):** "In life, there are events, both large and small, that end up affecting us. Think back over the past day and write down five events that somehow affected you."
- **Experimental Condition: Gratitude Letter (Walsh et al., 2022):** "In our daily lives, we often thank people for both large and small things. However, it can sometimes be difficult to make time to express our gratitude more deeply. Today, we want you to write a letter of gratitude to someone who has done something for which you are extremely grateful. Please do not share your letter with this person or anyone else."

Participants will also be told:

> As you write, don't worry about perfect grammar and spelling, and remember that anything you write will remain strictly confidential. Should an experimenter read this entry in the future, it will be identifiable only by a participant number and not by your name'. (Dang et al., 2025).

Some potential challenges of replicating this study will be ensuring that participants actually complete the task, especially since data collection will occur online without supervision. To mitigate this, we will use `jsPsych` (de Leeuw et al., 2023) and prevent participants from using copy and paste functions. Of interesting note, the original Dang et al. (2025) study did not exclude any participants from their analyses which reflects an intent-to-treat analysis approach (Ahn & Kang, 2023) which analyzes the data under unideal circumstances including non-compliance to reflect real-world treatment effects. In the interest of strict replication, I will follow the same procedure. I also think this is more interesting because often times, these positive psychology interventions are aimed at a large scale population (e.g., public school administrators recommending these strategies to students). Thus, non-compliance due to the nature of the task actually is inseparable from the outcome associated with the intervention. Simply analyzing the treatment effect under ideal conditions will hide these potential outcomes.

The key test in this study is a independent samples t-test. The Hedges' g will be a follow-up effect size calculation.

Reference List

Ahn, E., & Kang, H. (2023). Intention-to-treat versus as-treated versus per-protocol approaches to analysis. *Korean Journal of Anesthesiology*, *76*(6), 531–539. https://doi.org/10.4097/kja.23278

Cunha, L. F., Pellanda, L. C., & Reppold, C. T. (2019). Positive psychology and gratitude interventions: A randomized clinical trial. *Frontiers in Psychology*, *10*. https://doi.org/10.3389/fpsyg.2019.00584

Dang, A. V., Coles, N. A., Oishi, S., & McCullough, M. E. f(2026). The efficacy of seven gratitude interventions for promoting subjective well-being. *The Journal of Positive Psychology*, *21*(3), 517–535. https://doi.org/10.1080/17439760.2025.2502483

de Leeuw, J. R., Gilbert, R. A., & Luchterhandt, B. (2023). jsPsych: Enabling an open-source collaborative ecosystem of behavioral experiments. *Journal of Open Source Software*, *8*(85), Article 5351, https://joss.theoj.org/papers/10.21105/joss.05351.

Emmons, R. A., & McCullough, M. E. (2003). Counting blessings versus burdens: An experimental investigation of gratitude and subjective well-being in daily life. *Journal of Personality & Social Psychology*, *84*(2), 377–389. https://doi.org/10.1037/0022-3514.84.2.377

Walsh, L. C., Regan, A., Twenge, J. M., & Lyubomirsky, S. (2022). What is the optimal way to give thanks? comparing the effects of gratitude expressed privately, one-to-one via text, or publicly on social media. *Affective Science*, *4*(1), 82–91. https://doi.org/10.1007/s42761-022-00150-5

Learning Goals

Through this project, I hope to learn how to use jsPsych and SONA for data collection as well as get more familiarity with using Quarto to write academic reports. I also want to learn how to use R more (which I have begun to get better at since I have used it already in the power section). I think that this is a feasible project assuming that SONA has a sufficient amount of students because I have done similar projects in the past in undergrad (with less participants and no coding of the stimuli or data analysis) and those were completed quite reasonably fast.

I also hope this project will give me another experience with pre-registering hypotheses and making my analysis scripts/data open (I will completely redact people's lists and letters that they write, though).

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Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

The key effect size for this replication (gratitude letter vs. events list) is Hedges' $g = .41$. Many power analysis programs only offer Cohen's d power estimates. With larger samples, the difference between Hedges' g and Cohen's d becomes more minimal as the $J(x)$ correction factor applied to Cohen's d approaches 1 with large n.

Using the Hedges' g to Cohen's d conversion, we can divide the $J(x)$ correction factor to recover Cohen's d where $x = n_1 + n_2 - 2$. In Study 1, both conditions had exactly $n = 126$, therefore $x = 126 + 126 - 2 = 250$.

$$
\begin{aligned}
J(x) \cdot d &= g \\
d &= \frac{g}{J(x)} \\
d &= \frac{g}{\frac{\Gamma(x/2)}{\sqrt{x/2} \times \Gamma((x-1)/2)}}
\end{aligned}
$$

$$
\begin{aligned}
&\#\text{Substitute J(X)} = \text{J(250)} \\
d &= g \times \frac{\sqrt{250/2} \times \Gamma((250-1)/2)}{\Gamma(250/2)} \\
d &= g \times \frac{\sqrt{125} \times \Gamma(124.5)}{\Gamma(125)} \\
d &= g \times 1.003013...
\end{aligned}
$$

```
# Hedges' g Multiplied Against J(250)
GToCohensD = 0.41 * (sqrt(125)*gamma(124.5))/(gamma(125))
```

```
options(repos = c(CRAN = "https://cloud.r-project.org"))
install.packages("pwr")
```

```
The downloaded binary packages are in
    /var/folders/0p/p3phj575333gmy1dhsm2fsqr0000gp/T//RtmpO0o9qO/downloaded_packages
```

```r
library(pwr)
# 80% Power Needs 93.79 (94) participants per group for a two-sided test at alpha = .05. This
pwr.t.test(d = GToCohensD, sig.level = 0.05, power = 0.80, type = "two.sample", alternative =
```

```
     Two-sample t test power calculation

              n = 93.79321
              d = 0.4112351
      sig.level = 0.05
          power = 0.8
    alternative = two.sided

NOTE: n is number in *each* group
```

```r
# At 90% power, you need about 125 participants per group (total  = 250)
pwr.t.test(d = GToCohensD, sig.level = 0.05, power =.90, type = "two.sample", alternative = '
```

```
     Two-sample t test power calculation

              n = 125.2323
              d = 0.4112351
      sig.level = 0.05
          power = 0.9
    alternative = two.sided

NOTE: n is number in *each* group
```

```r
# At 90% power, you need about 155 participants per group (total  = 310)
pwr.t.test(d = GToCohensD, sig.level = 0.05, power =.95, type = "two.sample", alternative = '
```

```
     Two-sample t test power calculation

              n = 154.6466
              d = 0.4112351
      sig.level = 0.05
          power = 0.95
    alternative = two.sided

NOTE: n is number in *each* group
```

Planned Sample

Aligned with the original study, we will analyze as an intent to treat and therefore not exclude participants. Aiming for 80% power, we will aim for 188 total participants with complete data. This should balance power and feasibility of collecting data.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.f